

Climate Shocks, Health Finance, and Economic Outcomes

County-Level Evidence from the United States, 2011–2023

Thesis Committee Update · April 2026

Project Background

Climate, health finance, and economic outcomes in the U.S., 2011–2023

Research Question

- Do climate shocks (cold, heat, drought, air quality) affect the **cost and accessibility** of health care?
- Do they propagate into **household economic outcomes** — income, debt, employment?

Motivation

- Climate adaptation policy needs **health-finance channel** evidence
- Literature centres on mortality; **cost & debt channels understudied**
- U.S. county two-way FE enables within-unit identification

Data Coverage

Domain	Source
Climate (Temp, HDD, CDD)	NOAA/NCEI
Drought index (PDSI)	NOAA/NCEI
Air quality (AQI)	EPA
HIX premiums	HIX Compare
Medical debt	Urban Institute
Hospital bad debt	NASHP HCT
Employer insurance	AHRQ MEPS-IC
Public payers	CMS NHE
HH income / employment	ACS / BEA

Panel: 2011–2023; ~3,155 counties;

state panel: 51 states × 13 years.

Empirical Specification

Two-way fixed effects with distributed lags; all dollar outcomes inflation-adjusted to \$2023

Estimating Equation

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^2 \beta_h \text{Shock}_{i,t-h} + \mathbf{X}'_{it} \delta + \varepsilon_{it}$$

- α_i : unit FE (state or county); γ_t : year FE
- Distributed lags $h = 0, 1, 2$ for all shocks
- SE clustered at **state level** throughout
- **County**: also rating-area-clustered SEs for premiums

Climate Shock Variables

Variable	Definition
High_CDD	$CDD \geq \text{national p80 (1990–2000)}$
High_HDD	$HDD \geq \text{national p80 (1990–2000)}$
Z_Temp	$(T_{it} - \bar{T}_i^{90-00}) / \sigma_i^{90-00}$
Z_Precip	Precipitation z-score (same baseline)
PDSI	Palmer drought severity index
High_AQI_Max	Max AQI > 100 (binary)

Outcome Blocks

Outcome	Source
Employer premium contribution	MEPS-IC
Benchmark Silver premium	HIX
Medical debt share / median	Urban Inst.
Hospital bad debt per capita	NASHP
Total, Medicaid, Medicare spending	CMS
Median HH income / employment	ACS
Per-capita personal income	BEA

Dynamic Estimation Strategy

Baseline dynamic-panel distributed lags and local projections recover complementary timing margins

Dynamic Panel / Distributed Lag (DP/DL)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^2 \beta_h \text{Shock}_{i,t-h} + \mathbf{X}'_{it} \delta + \varepsilon_{it}$$

- Main FE estimator for contemporaneous and lagged effects
- Cumulative response is the sum of lag coefficients
- Baseline estimator for the state and county tables

Controls in both approaches: uninsured rate, median household income, and the same county/year fixed effects.

Inference: state-clustered SEs throughout; rating-area clustered variants for premium outcomes as robustness.

Local Projections (LP, Jordà 2005)

$$Y_{i,t+h} = \alpha_i^h + \gamma_t^h + \beta_h \text{Shock}_{i,t} + \mathbf{X}'_{it} \delta_h + \varepsilon_{i,t+h}$$

- Separate regression for each horizon $h \in \{-2, \dots, +3\}$
- Negative horizons serve as placebo leads for pre-trend checks
- More flexible when lag shapes are misspecified

Roadmap: Three Complementary Analyses

This presentation covers state, county, and a new weather-swing analysis

State-Level FE

- 51 states $\times$ 13 years
- Drought and cold channels on premiums and debt
- Establishes the aggregate picture

County-Level FE

- $\sim 3,155$ counties, 2011–2023
- HDD, CDD, drought, AQI on debt, premiums, income, hospital costs
- Higher power; rating-area clustering for premiums

Weather Swing (New)

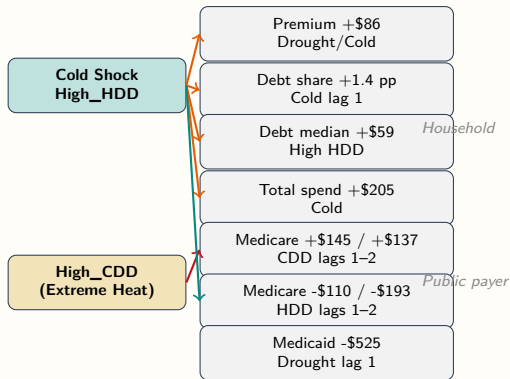
- County-level only
- Exposure:
 $\Delta X_{it} = X_{it} - X_{i,t-1}$
- Identifies cost of **volatility**, not just level
- Finds effects where level specs were insignificant (AQI)

State panel ($N \approx 650$) has insufficient power for first-difference specifications — swing analysis is county-only.

State-Level Results

Two-way FE: State + Year, $N \approx 550-650$, cluster = State

State-Level: Cold and Heat Drive Opposite Financial Channels



Cold channel:

- Premiums, debt, and total spending rise
- Medicare falls with lagged HDD exposure

Heat channel (CDD):

- Medicare rises 1–2 years later

Drought:

- Medicaid falls and premiums rise

Heat and cold affect public payers in opposite directions — novel finding.

State: Drought and Cold Raise Premiums and Debt

Household & individual outcomes; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Employer Premium Contribution (\$/yr)

Predictor	Coef.	
Extreme Drought ($h=0$)	+63.6	**
Severe Drought ($h=0$)	+86.4	***
Severe Drought (lag 1)	+65.1	**
Pct PM _{2.5} days	+20.8	***
Pct PM ₁₀ days	+22.7	***
Pct Ozone days	+19.7	***

Air

quality pollutant days show consistent positive effects on employer premium costs.

Medical Debt Share (pp)

Predictor	Coef.	
Cold Shock (lag 1)	+0.0135	**

Medical Debt Median (\$/yr)

Predictor	Coef.	
High HDD ($h=0$)	+58.9	**
Severe Drought ($h=0$)	-66.8	***
Severe Drought (lag 1)	-37.5	**
Heat Shock ($h=0$)	+32.7	*
Heat Shock (lag 2)	+47.7	**

Cold raises debt prevalence; drought paradoxically reduces median severity.

State: Cold Raises System Costs; Drought Cuts Public Payer Spending

Hospital & system outcomes; two-way FE (State + Year)

Total Per-Capita Health Expenditure (\$)

Predictor	Coef.	
Cold Shock ($h=0$)	+205.3	**
High HDD (lag 2)	+92.0	*

Medicaid per Enrollee (\$)

Predictor	Coef.	
Extreme Drought ($h=0$)	-566.2	**
Severe Drought ($h=0$)	-525.8	**
Severe Drought (lag 1)	-524.5	***
Severe Drought (lag 2)	-288.4	**

persistently reduces Medicaid
spending across lags 0–2.

Drought

Medicare per Enrollee (\$)

Predictor	Coef.	
Severe Drought (lag 1)	-144.3	***
High CDD (lag 1)	+144.7	**
High CDD (lag 2)	+137.2	***
High HDD (lag 1)	-110.1	*
High HDD (lag 2)	-192.7	***
Unhealthy days	+97.7	**

Heat

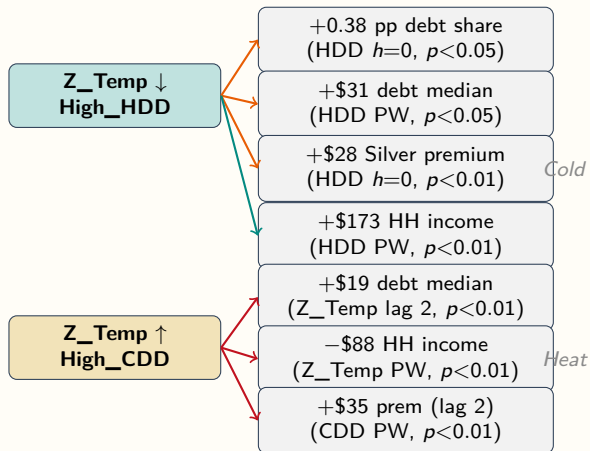
(CDD) raises Medicare with lag;
cold (HDD) reduces it — opposite channels.

County Results: Household & Individual

Medical debt, premiums, income, employment

Two-way FE: County + Year, $N \approx 23k-34k$, cluster = State

County-Level: Cold and Heat Channels Mirror the State Findings



Cold channel:

- Raises debt prevalence and severity
- Raises premiums on impact
- Raises HH income (energy mix?)

Heat channel:

- Raises debt median (lagged)
- Reduces HH income contemporaneously
- Raises premiums with 1–2 yr lag

Cross-level:

- Cold → debt: both levels confirmed
- Heat → income: county only
- Premium channels align in direction

County: Heat Raises Debt Severity; Cold Raises Debt Prevalence

Medical Debt Share (pp) and Median Debt (\$2023); UW = unweighted, PW = pop-weighted

Z_Temp (Relative Heat Shock)

	UW	PW	
<i>Debt Share (pp)</i>			
<i>h=0</i>	−0.0025*	−0.0000	Heat
Lag 2	−0.0027**	−0.0014*	
<i>Debt Median (\$)</i>			
<i>h=0</i>	+14.9*	+18.5***	
Lag 2	+19.5**	+16.2***	

reduces prevalence but raises severity
— consistent composition effect:
warm years shed marginal debtors while
pushing the remaining cases deeper.

High_HDD (Extreme Cold Burden)

	UW	PW
<i>Debt Share (pp)</i>		
<i>h=0</i>	+0.0038**	+0.0054**
<i>Lag 2</i>	+0.0031*	+0.0037*
<i>Debt Median (\$)</i>		
<i>h=0</i>	+10.6	+31.2**

Controls (all debt models):

Uninsured Rate +*** (large)
HH Income —**

Cold raises both prevalence and severity
— no composition effect.

County: Cold Raises Premiums Now; Heat Raises Them Later

Benchmark Silver Premium (\$2023 real); county + year FE

High_HDD (Cold Burden)

Lag	UW	PW	Cluster
$h=0$	+28.1***	+21.9**	State
$h=0$	+28.1***	+21.9***	RA
Lag 1	-24.5*	-20.8	State

High_CDD (Heat Burden) — lagged

Lag	UW	PW	Cluster
Lag 1	+8.7	+31.4***	RA
Lag 2	+21.6*	+35.3***	RA

Z_Temp (Relative Temp)

Lag	UW	PW	Cluster
$h=0$	-10.9**	+2.7	RA
Lag 1	+3.8	+5.4**	RA
Lag 2	+1.0	+4.2**	RA

- **Cold (HDD)** raises premiums on impact in both state- and RA-clustered specs
- **Heat** lowers premiums now, then raises them 1–2 years later
- RA clustering tightens CIs; drought lag 1 is borderline at +\$3.9 ($p=0.05$)

County: Heat Depresses Income; Drought Hits PCPI

Median HH income and per-capita personal income (\$2023)

Median HH Income (\$2023)

Predictor	UW	PW	Relative
Z_Temp ($h=0$)	-71**	-88***	
Z_Temp (lag 1)	-36	-64***	
Z_Temp (lag 2)	-71**	-60**	
High_CDD ($h=0$)	-247**	-3	
High_HDD ($h=0$)	+161**	+173***	

heat most consistently lowers household income in the population-weighted specifications.

Per-Capita Personal Income (\$)

Predictor	UW	PW
PDSI lag 2	-8.5	-113**
High_CDD lag 1	+480**	-116
RA-level: PDSI ($h=0$)	-206***	

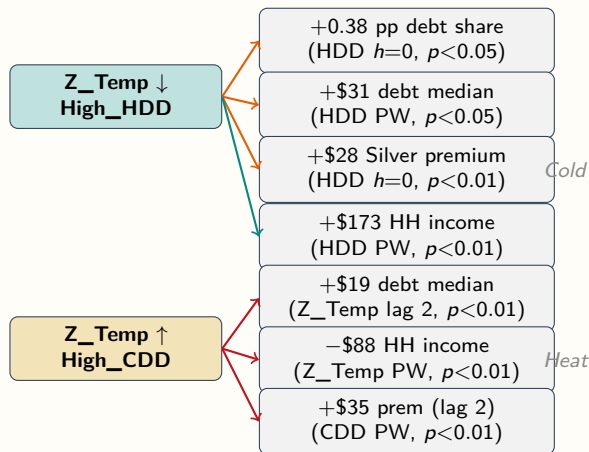
Income Summary

- **Heat** lowers HH income across lags in pop-weighted models
- **Cold** raises HH income on impact, likely reflecting seasonal or energy-income composition
- **Drought** is strongest in PCPI: RA spec gives -\$206***

Civilian Employment (persons)

Predictor	UW	PW
Z_Temp lag 2	+788**	+3216
High_HDD ($h=0$)	-468**	+2587
High_HDD lag 2	-678**	+3988
Z_Precip ($h=0$)	+46	-5742***
Z_Precip (lag 1)	-88	-3803***

County-Level: Cold and Heat Channels Mirror the State Findings



Cold channel:

- Raises debt prevalence and severity
- Raises premiums on impact
- Raises HH income (energy mix?)

Heat channel:

- Raises debt median (lagged)
- Reduces HH income contemporaneously
- Raises premiums with 1–2 yr lag

Cross-level:

- Cold → debt: both levels confirmed
- Heat → income: county only
- Premium channels align in direction

County Results: Hospital Sector

Hospital bad debt per capita

Two-way FE: County + Year, $N \approx 27k$, cluster = State

County: Cold Raises Hospital Bad Debt; Heat Lag Reduces It

Hospital Bad Debt per Capita (\$); within- $R^2 = 0.006$ – 0.019

Key Coefficient Estimates

Predictor	UW	PW
Z_Temp lag 2	−2.41***	−2.08***
Z_Temp ($h=0$)	−1.25	−1.08**
High_HDD ($h=0$)	+5.33***	+4.23**
PDSI lag 1	+0.57*	+0.50*
PDSI lag 2	+0.01	+0.50*
Uninsured Rate	+151***	+142

RA-level robustness (pop-wtd):

PDSI lag 2: +0.76*** Z_Temp lag 2: −2.26***

- **Cold (HDD) raises bad debt on impact** by \$4–5 per capita — consistent with emergency utilisation, deferred billing, and coverage gaps during cold events
- **Heat lag reduces bad debt** (2-yr lag, $p < 0.001$) — possible reduction in elective utilisation during hot years or survivor selection
- **PDSI (drought)** adds a smaller lagged increase, confirmed in RA robustness
- **Uninsured Rate** is the dominant control (+151*** unweighted) — structural coverage gap drives hospital write-offs

About 23% of county-years lack hospital reports; one negative outlier is winsorized.

Weather Swing Analysis

A new source of variation: year-over-year climate volatility
County + Year FE, $N \approx 23k-34k$, cluster = State

New Analysis: Exploiting Year-to-Year Weather Swings

County-level only; $N \approx 650$ state-years insufficient for first-difference identification

Motivation

- Prior specs identify counties in persistent shock states vs. not
- New margin: do counties that **swing sharply** year-to-year bear extra health costs, regardless of average climate?
- Is it **volatility**, not just level, that damages health finance?

Robustness Specs

- **Asymmetric:** $\Delta^+ = \max(\Delta X, 0)$ vs. $\Delta^- = \min(\Delta X, 0)$
- **Binary transitions:** Onset / Exit / Persist indicators

Primary Specification

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 \underbrace{\Delta X_{it}}_{\text{swing}} + \beta_2 \underbrace{X_{i,t-1}}_{\text{lagged level}} + \mathbf{X}'_{it} \delta + \varepsilon_{it}$$

- $\Delta X_{it} = X_{it} - X_{i,t-1}$ for all continuous exposures:
Z_Temp, Z_Precip, CDD, HDD, PDSI, Median_AQI, Max_AQI
- β_2 controls for prior-year level — isolates pure swing effect
- LP extension: $Y_{i,t+h}$ at $h = 0, 1, 2, 3$
- County + year FE; state-clustered SEs

Weather Swings Affect Premiums, Debt, and Hospital Costs

Primary ΔFE spec ($h=0$, unweighted); * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

$\Delta HDD \rightarrow$ Silver Premium

Spec	Estimate	p	
ΔHDD ($h=0$)	+0.033***	0.0004	Cold
HDD Onset (binary)	+49.46**	0.0033	
HDD_Pos (asymm.)	+0.065*	0.028	

swing raises premiums on impact;
onset of first extreme-cold year
adds $\sim$ \$49 to Silver benchmark.

$\Delta Z_Temp \rightarrow$ Medical Debt Median

Horizon	Estimate (\$)	p	
$h=0$	+13.2*	0.014	Warming swing
$h=2$	+19.4*	0.033	
$h=3$	+29.7***	0.0001	

raises median debt
with a growing 3-year lag.

$\Delta Max_AQI \rightarrow$ Hospital Bad Debt (per capita)

Horizon	Estimate (\$)	p	
$h=0$	+0.0050**	0.0014	AQI swing effect
$h=1$	+0.0082***	<0.001	
$h=2$	+0.0080*	0.015	
$h=3$	+0.0157***	<0.001	

on bad debt **grows** across
all four horizons — a persistent ratchet.
Level effect was not significant in Phase 2.

$\Delta Z_Precip / \Delta PDSI \rightarrow PCPI$

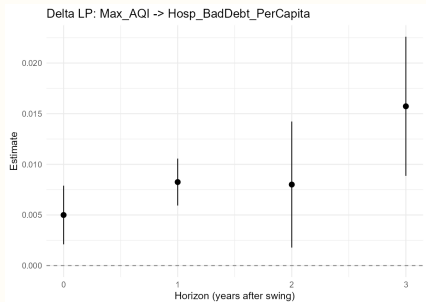
Exposure	$h=1$ est.	p	
ΔZ_Precip	-274***	0.0006	Precipitation and
$\Delta PDSI$	-226***	0.0001	

drought swings
suppress per-capita income for 1–3 yrs.

Swing Effects Grow Over Time: Two Key LP Profiles

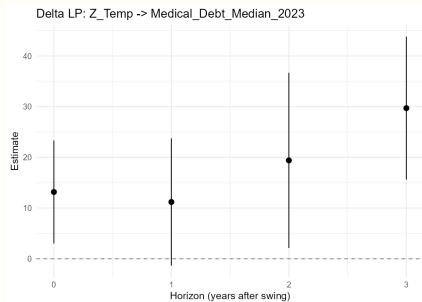
Local projection estimates at $h = 0, 1, 2, 3$; county + year FE; state-clustered SEs; unweighted

$\Delta \text{Max_AQI} \rightarrow \text{Hospital Bad Debt}$ (\$/capita)



Effect grows at every horizon — AQI bad-debt ratchet. Level effect was not significant in Phase 2.

$\Delta Z_Temp \rightarrow \text{Median Medical Debt}$ (\$2023)



Warming swing accumulates into medical debt; +\$30 at $h=3$ ($p < 0.001$).

Asymmetry and Ratchets: Not All Swings Are Reversible

Asymmetric spec (Δ^+ vs. Δ^-) and onset/exit indicators

Asymmetric Findings

- **AQI ratchet:** improving air quality still raises bad debt ($\Delta^-, +0.010^{***}$) — costs do not unwind when pollution falls
- **Drought worsening:** Δ^- PDSI drives PCPI loss (-207^{***}); recovery asymmetrically weaker
- **CDD income:** symmetric — cooling-sector PCPI moves with swings in both directions ($\pm 3.3, p < 0.001$)
- **Drought exit** *does* reduce medical debt ($-\$54^*$)

What the Delta Analysis Adds

Relationship	Finding
AQI $\rightarrow$ hosp. bad debt	New: swing sig.; level not
HDD $\rightarrow$ premiums	Confirms; adds onset pricing
Drought $\rightarrow$ income	Confirms; worsening-only
Temp swing $\rightarrow$ med. debt	New: 3-yr growing
Precip swing $\rightarrow$ PCPI	New: 3-yr suppression

Key insight: weather volatility carries independent health-finance costs beyond level effects.

Cross-Level Synthesis

What do state, county, and swing analyses agree on?

State and County Pipelines Tell a Consistent Story

Key findings replicated across both levels of analysis

Finding	State	County
Cold / HDD → medical debt (prevalence & severity)	✓	✓
Cold / HDD → higher premiums	✓ (via drought)	✓ ($p<0.01$)
Cold / HDD → hospital bad debt ↑	—	✓ ($p<0.01$)
Drought → reduced Medicaid / PCPI	✓ ($p<0.01$)	✓ (PDSI)
Heat → household income loss	—	✓ (Z_Temp***)
Heat lag → Medicare / premiums ↑	✓ (CDD***)	✓ (CDD***)
Heat lag → hospital bad debt ↓	—	✓ ($p<0.001$)
Poor air quality → higher premiums	✓ (PM _{2.5} /O ₃ ***)	not modeled

Next Steps

Planned extensions to the empirical framework

Empirical Extensions

- **Heterogeneity:** rural vs. urban and high-vs. low-AQI subsamples for both level and swing analyses
- **Mechanisms:** separate respiratory-illness from energy-poverty channels for cold shocks
- **County unemployment:** replace the ACS proxy with BLS LAUS series
- **Policy counterfactuals:** ACA Medicaid expansion $\times$ climate shocks

Writing and Defence

- **Chapter draft:** unify state, county, and swing narratives into a single empirical chapter
- **Swing robustness:** population-weighted LP profiles for delta results; cumulative IRF plots
- **Thesis write-up:** convert synthesis tables into publication-ready regression tables